



OmniShaperAI

You imagine
We create



Problems of 3D modelling field

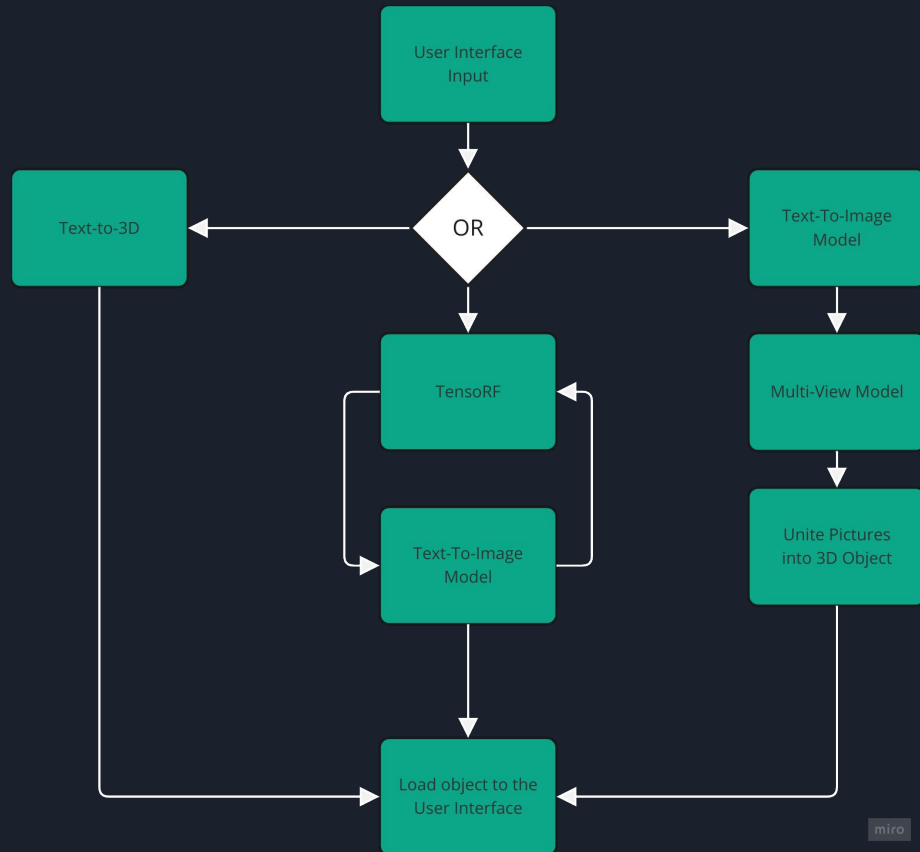
- Game developers **who have zero knowledge** about 3D modeling and focus on writing scripts and game logic that causes them to use already existing and paid assets that have limited capabilities
- **Professional 3D modelers** that work in game development and architecture fields have to deal with a daily routine of creating “base” 3D models for more complex ones



Our solution

- We implemented a [web-service](#) which provides the user with a high-quality Text-to-3D AI model
- The user may choose one of the 2D pictures generated by the diffusion model - the [base image](#)

How the pipeline works





The first idea

After studying the materials on [fresh models](#) (Text-to-2D, 2D-to-3D, 2D-to-3D), there were 3 primary ideas:

- [Direct Text-to-3D generation](#)
There were no models to implement
- [Stable Diffusion V1.5 → TensorRF](#)
There was a problem with running TensorRF, therefore, we did not succeed with this idea.
- [Stable Diffusion V1.5 → SyncDreamer → TensorRF](#)
The obstacle faced this time was figuring out how to properly convert necessary information about views into a rotation matrices.
- [Stable Diffusion V1.5 → Wonder3D → NeRF](#)
Wonder3D model gave an ability to generate new views with the properly initialized rotation matrices.



The overall progress

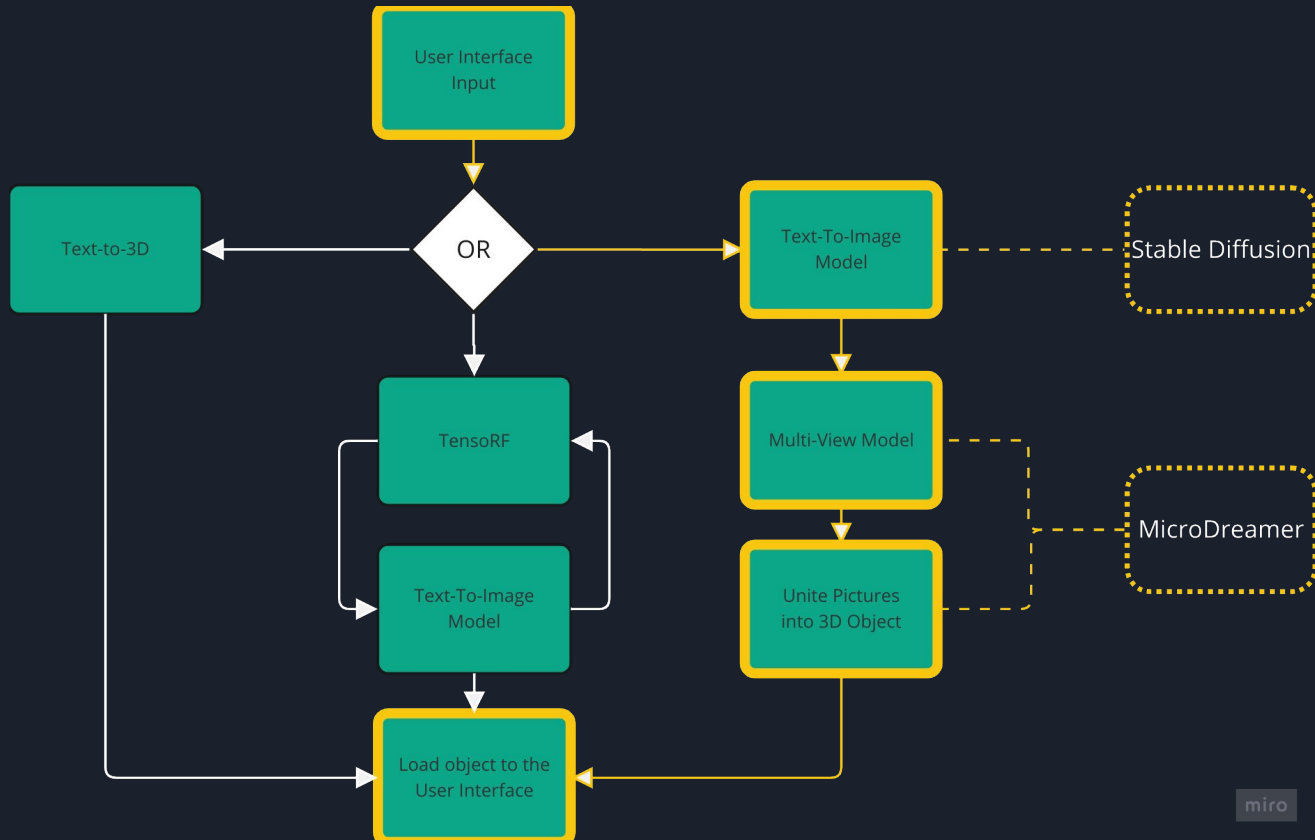
Quite a lot of time was spent to find the proper architecture and implement it. Such iterating through different ideas gave us the best performance we could reach.

For now, our idea became the state of art.

We decided to stop on:

- [Stable Diffusion 3](#)
- [MicroDreamer](#)

How the pipeline works





Demonstration part

Now we are going to show you the progress we made on the quality of our models.



Thank you for your attention